

Verify Important Outputs

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Repair a unit error for Verifying claims and quantities. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Verifying claims and quantities. Proposed explanation: “A citation proves every nearby claim”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a new model with mass units

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss. A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four claims, show the remaining mass in grams, and explain the status of the prize claim. Use $1 \text{ kg} = 1000 \text{ g}$.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A generated report says 18 of 24 tests passed, so success was 80%. Verify the claim in two ways and write a corrected sentence.

6. Solve the given case

An answer says six groups of four make 28.

Verify independently.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An assistant says the school library opens at 8, but today's official notice says 9. Which source should guide today's visit? Explain the check.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 27 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Require human confirmation for important sorter outputs and attach the evidence record used to approve each rule or label.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A generated report says 18 of 24 tests passed, so success was 80%. Verify the claim in two ways and write a corrected sentence.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Separate source facts from derived quantities

Using R, identify a directly stated fact and a quantity that must be calculated. Give the source or expression for each. Explain why a citation to R does not verify the claim that every participant will enjoy the activity.

My reasoning and evidence

13. W2 Repair a unit error

A report calculates $12 \times 50 = 600$ and labels the result metres. Use R to correct the unit, convert the result to metres and calculate the remaining length. Give a second arithmetic route or a check using starting, used and remaining lengths.

My reasoning and evidence

14. W3 Do not count copied claims as separate evidence

Three posters say 600 metres were used, but all copied one draft. A fourth poster cites card Z, which is unavailable. Explain why these posters do not settle the quantity. Name the available evidence and check you would use instead.

My reasoning and evidence

15. W4 Audit a new model with mass units

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss. A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four claims, show the remaining mass in grams, and explain the status of the prize claim. Use 1 kg = 1000 g.

My reasoning and evidence

16. W5 State what a missing source prevents you from knowing

Another draft claims that the amount left in W4 accounts for all real-world waste, citing a missing instruction card. Explain which result can still be checked from M and which claim remains unresolved. What information would be needed before revising the no-loss model?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Verify Important Outputs

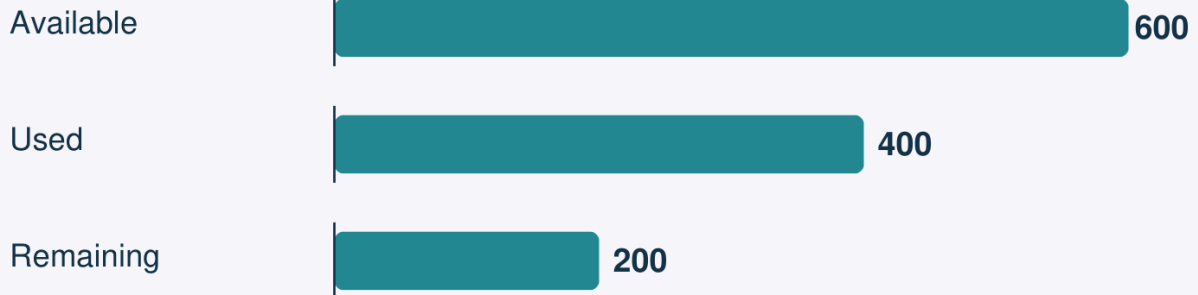
A fictional craft sheet calculates ribbon left after making bookmarks.

Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

Ribbon quantities in one consistent unit

centimetres



Bar length shows the supplied value; the marked baseline is zero.

Verify Important Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Available ribbon

$$4 \times 150 = 600 \text{ cm.}$$



2. Ribbon used

$$8 \times 50 = 400 \text{ cm.}$$



3. Remainder

$$600 - 400 = 200 \text{ cm} = 2 \text{ m.}$$



4. Audit draft

"200 m remains and everyone wins" fails unit and evidence checks.

CHANGE ONE THING

Make ten bookmarks instead of eight.

Explain the changed result using the matching lesson.

Correct the draft and identify which part needs another source rather than a calculation.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Check sorter calculations with explicit units and verify explanation claims against their actual catalogue evidence.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Check sorter calculations with explicit units and verify explanation claims against their actual catalogue evidence.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft sheet calculates ribbon left after making bookmarks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? “200 m remains and everyone wins” fails unit and evidence checks.

1. Audit draft
2. Remainder
3. Ribbon used
4. Available ribbon

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails unit and evidence checks. All arithmetic first uses centimetres. Converting only after subtraction avoids mixing units. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning. A numerically correct total elsewhere in a draft cannot rescue a wrong unit or an invented claim.

Practice 2: explain the deciding evidence

Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. 100 cm, or 1 m, remains.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains. $600 - 10 \times 50 = 100$ cm.

Repair a missing building block

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied evidence and checkable output requirements. Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields. Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are evidence-backed in the original case, and which row must remain unknown?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
